



Lab Report – EE-170L Computer Programming

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DATE	26/06/2026

COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB 9

Lab Title: Introduction to Functions in C++

1. OBJECTIVE

The objectives of this lab are:

- To understand what a function is and why functions are used in C++.
- To learn how to declare, define, and call user-defined functions.
- To understand function prototypes and the role of parameters and return types.
- To implement programs that use functions to organize and reuse code.

2. THEORY

Introduction to Functions

Functions let you divide a long program into named sections so that those sections can be reused throughout the program. A function has a name identifier, accepts parameters, and returns a result. C++ functions can accept an unlimited number of parameters. In general, C++ does not care in what order functions appear in a program, as long as the function name is known to the compiler before it is called.

We have been consistently using library functions such as `cout` and `cin` without knowing their internal implementation. We simply call them with arguments and they perform the required work, then return control to the calling program. In this lab we learn how to write our own functions that perform specific tasks.

A function is distinguished from a variable by the parentheses that follow its name, for example `main()`. Anything inside those parentheses are the parameters passed to the function. In the case of multiple parameters, each is separated by a comma. If no parameters are needed, the parentheses are left empty.

Prototype of a Function

A function prototype describes the function to the compiler before its full definition appears. It specifies the function name, parameter types, and return type, allowing the compiler to verify that function calls are correct.

Syntax:

```
returnType functionName(parameterType1, parameterType2, ...);
```

Examples:

```
int main(); float divide(int  
a, int b); void square(float  
x);
```

Here, the functions main, divide, and square take 0, 2, and 1 parameters respectively. A return type of void means the function returns nothing. A return type of int or float means the function returns a value of that type using the return statement.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. TASKS

Task 1 — Function Introduction and Declaration

A function named greet was declared with no parameters and no return value. Inside the function a greeting message is displayed. The function is called from main to execute the greeting.

```
#include <iostream> using  
namespace std;  
  
void greet(); // function prototype: no parameters, no return value  
  
int main()  
{  
    greet(); // call the function  
    return 0;  
}  
  
// function definition void  
greet()  
{  
    cout << "Welcome to Computer Programming Lab!" << endl;  
}
```

Task 2 — Basic Function Declaration and Call

A function named `sayHello` was declared with no parameters and no return value. Inside the function, the message "Hello, World!" is displayed. The function is called from the main program.

```
#include <iostream> using
namespace std;

void sayHello(); // function prototype

int main()
{
    sayHello(); // call the function to print the message
    return 0;
}

// function definition void
sayHello()
{
    cout << "Hello, World!" << endl; // display the greeting
}
```

Task 3 — Array Sum and Average Using a Function

The program reads 10 integer values from the user and stores them in a 1D array. A function named `calculateSum` accepts the array and its size as parameters and returns the total sum of all elements. The main function calls `calculateSum`, then divides the result by the number of elements to compute and display the average.

```
#include <iostream> using
namespace std;

// prototype: accepts array and size, returns sum int
calculateSum(int arr[], int size);

int main()
{
    int nums[10];

    // read 10 integers from the user into the array
    for (int i = 0; i < 10; i++)
    {
        cout << "Enter number " << i + 1 <<
        ": ";
        cin >> nums[i];
    }

    int total = calculateSum(nums, 10); // call the function
    double avg = (double)total / 10;    // compute average

    cout << "Sum      = " << total << endl;
    cout << "Average = " << avg  << endl;    return
    0;
}

// function definition: sums all elements of the array int
calculateSum(int arr[], int size)
{
    int sum = 0;    for (int i =
    0; i < size; i++)
    {
        sum += arr[i]; // add each element to the running
total
    }
    return sum; // return the final sum to the caller
}
```

5. CONCLUSION

This lab introduced user-defined functions in C++. The structure of a function was examined through its prototype, definition, and call. Functions with no parameters, functions that return values, and functions that accept arrays were all implemented. Using functions to separate logic into named, reusable blocks makes programs easier to read, test, and maintain, and this principle will be applied throughout the remainder of the course.